Convergence of the Sequence $u_n = \sqrt{2 + u_{n-1}}$

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1 Introduction

This document presents a problem found in **HSC Mathematics Extension** course, Australian Curriculum, Year 12. The problem involves analyzing the convergence of a recursively defined sequence using fundamental theorems from calculus and real analysis. This topic is essential for students studying advanced sequences and series, as it demonstrates the powerful application of the Monotone Convergence Theorem.

We will explore how sequences defined by recurrence relations behave, and learn to prove their convergence rigorously. The techniques used here are fundamental tools that you will encounter throughout your mathematical studies.

2 Problem Statement

Consider the sequence (u_n) defined recursively by:

$$u_n = \sqrt{2 + u_{n-1}}, \quad n \geq 1 \tag{1}$$

with an initial value $u_0 > 0$.

Question: Does this sequence converge? If so, what is its limit, and how can we prove this convergence?

This problem challenges us to:

i) Determine the potential limit (if it exists)

ii) Find the closed-form solution for u_n

iii) Prove that the sequence actually converges to this limit

iv) Understand the behavior of the sequence for different starting values

Flip to the next page for the solution.

3 Solution

To discuss the convergence of the sequence $u_n = \sqrt{2 + u_{n-1}}$ with $u_0 > 0$, we will use the **Monotone Convergence Theorem**, which states that if a sequence is both monotonic (either increasing or decreasing) and bounded, it must converge.

Monotone Convergence Theorem

If a sequence (a_n) is monotonic (either increasing or decreasing) and bounded, then (a_n) converges.

Main Result: The sequence (u_n) **converges to 2** for any initial value $u_0 > 0$.

3.1 Finding the Potential Limit

First, let's assume the sequence converges to a limit L. If $u_n \to L$ as $n \to \infty$, then $u_{n-1} \to L$ as well. We can find the value of L by substituting it into the recurrence relation:

$$L = \sqrt{2 + L}$$

To solve for L, we square both sides (noting that L must be non-negative, as u_n is the result of a principal square root for all $n \geq 1$):

$$\begin{aligned} L^2 &= 2 + L \\ L^2 - L - 2 &= 0 \\ (L-2)(L+1) &= 0 \end{aligned}$$

This gives two possible limits: $L = 2$ or $L = -1$.

Since $u_0 > 0$, we have $u_1 = \sqrt{2 + u_0} > 0$. By induction, every term u_n must be positive. Therefore, the limit L must be non-negative.

Conclusion: The only possible limit for the sequence is $L = 2$.

3.2 Proving Convergence (Monotonicity and Boundedness)

Now we must prove that the sequence *does* converge. We analyze the behavior of the sequence based on the starting value u_0.

To determine if the sequence is increasing or decreasing, we examine when $u_n > u_{n-1}$:

$$\begin{aligned} \sqrt{2 + u_{n-1}} &> u_{n-1} \\ 2 + u_{n-1} &> u_{n-1}^2 \quad (\text{since } u_{n-1} > 0) \\ 0 &> u_{n-1}^2 - u_{n-1} - 2 \\ 0 &> (u_{n-1} - 2)(u_{n-1} + 1) \end{aligned}$$

Since $u_{n-1} > 0$, the term $(u_{n-1} + 1)$ is always positive. The inequality simplifies to:

$$0 > u_{n-1} - 2$$

which means $u_{n-1} < 2$.
This tells us:

- If $u_{n-1} < 2$, then $u_n > u_{n-1}$ (the sequence is **increasing**).
- If $u_{n-1} > 2$, then $u_n < u_{n-1}$ (the sequence is **decreasing**).
- If $u_{n-1} = 2$, then $u_n = 2$ (the sequence is **constant**).

We can now analyze the convergence by cases.

3.2.1 Case 1: $u_0 = 2$

If $u_0 = 2$, then $u_1 = \sqrt{2+2} = 2$. By induction, $u_n = 2$ for all n.
Conclusion: The sequence is constant and **converges to 2**.

3.2.2 Case 2: $0 < u_0 < 2$

a) **Monotonicity:** We know that if $u_{n-1} < 2$, the sequence increases. Let's prove by induction that u_n *stays* below 2.

   - **Base Case:** $u_0 < 2$ (given).
   - **Inductive Step:** Assume $u_k < 2$. Then

$$u_{k+1} = \sqrt{2+u_k} < \sqrt{2+2} = \sqrt{4} = 2.$$

   - Thus, $u_n < 2$ for all n.
   - Because $u_n < 2$ for all n, it follows from our earlier analysis that $u_{n+1} > u_n$ for all n. The sequence is **strictly increasing**.

b) **Boundedness:** We just proved by induction that $u_n < 2$ for all n. The sequence is **bounded above by 2**.

Conclusion (Case 2): The sequence is increasing and bounded above. By the Monotone Convergence Theorem, it converges. As shown in Step 1, the only possible limit is 2.

3.2.3 Case 3: $u_0 > 2$

a) **Monotonicity:** We know that if $u_{n-1} > 2$, the sequence decreases. Let's prove by induction that u_n *stays* above 2.

   - **Base Case:** $u_0 > 2$ (given).
   - **Inductive Step:** Assume $u_k > 2$. Then

$$u_{k+1} = \sqrt{2+u_k} > \sqrt{2+2} = \sqrt{4} = 2.$$

   - Thus, $u_n > 2$ for all n.
   - Because $u_n > 2$ for all n, it follows from our earlier analysis that $u_{n+1} < u_n$ for all n. The sequence is **strictly decreasing**.

b) **Boundedness:** We just proved by induction that $u_n > 2$ for all n. The sequence is **bounded below by 2**.

Conclusion (Case 3): The sequence is decreasing and bounded below. By the Monotone Convergence Theorem, it converges. As shown in Step 1, the only possible limit is 2.

3.3 Summary

In all possible cases for $u_0 > 0$, the sequence (u_n) is monotonic and bounded, and therefore **it always converges to the limit 2**.

This result demonstrates the power of the Monotone Convergence Theorem: by simply showing that a sequence is monotonic and bounded, we can guarantee its convergence without needing to compute the limit directly from the recurrence relation.

4 Finding Closed Forms

While the Monotone Convergence Theorem proves convergence, we can also find explicit closed-form expressions for u_n by treating the recurrence relation as a difference equation. This approach uses trigonometric and hyperbolic substitutions to derive exact formulas.

4.1 Difference Equation Approach: Closed-Form Solutions

The recurrence relation $u_n = \sqrt{2 + u_{n-1}}$ is a **non-linear first-order difference equation**. We can find closed-form solutions using clever substitutions based on trigonometric and hyperbolic identities. The form $\sqrt{2 + ...}$ suggests using the half-angle identities.

4.1.1 Case 1: $0 < u_0 \leq 2$

We use the substitution $u_n = 2\cos(\theta_n)$. The half-angle identity for cosine is $\cos(x/2) = \sqrt{\frac{1+\cos(x)}{2}}$, which can be rewritten as $2\cos(x/2) = \sqrt{2 + 2\cos(x)}$.

1. **Substitute:**

$$\begin{aligned} u_n &= \sqrt{2 + u_{n-1}} \\ 2\cos(\theta_n) &= \sqrt{2 + 2\cos(\theta_{n-1})} \end{aligned}$$

2. **Apply Identity:**

$$\begin{aligned} 2\cos(\theta_n) &= \sqrt{2(1 + \cos(\theta_{n-1}))} \\ 2\cos(\theta_n) &= \sqrt{2(2\cos^2(\theta_{n-1}/2))} \\ 2\cos(\theta_n) &= \sqrt{4\cos^2(\theta_{n-1}/2)} \\ 2\cos(\theta_n) &= 2|\cos(\theta_{n-1}/2)| \end{aligned}$$

3. **Solve for θ_n:** Since $0 < u_0 \leq 2$, we can set $u_0 = 2\cos(\theta_0)$ for some $\theta_0 \in [0, \pi/2)$. In this interval, cos is non-negative, so we can drop the absolute value.

This gives $\theta_n = \theta_{n-1}/2$, which is a simple geometric progression. The solution is:

$$\theta_n = \frac{\theta_0}{2^n}$$

4. **Find the Closed-Form Solution:** From $u_0 = 2\cos(\theta_0)$, we have $\theta_0 = \arccos(u_0/2)$. Substituting back, the closed-form solution for u_n is:

$$u_n = 2\cos\left(\frac{\arccos(u_0/2)}{2^n}\right)$$

5. **Discuss Convergence:** As $n \to \infty$, the argument of cosine goes to zero:

$$\lim_{n\to\infty}\left(\frac{\arccos(u_0/2)}{2^n}\right) = 0$$

Therefore, the limit of u_n is:

$$\lim_{n\to\infty} u_n = 2\cos(0) = 2 \cdot 1 = 2$$

4.1.2 Case 2: $u_0 > 2$

The substitution $u_n = 2\cos(\theta_n)$ fails because $u_0/2 > 1$, which is outside the domain of arccos. We use the hyperbolic cosine equivalent: $u_n = 2\cosh(\theta_n)$. The identity is $2\cosh(x/2) = \sqrt{2 + 2\cosh(x)}$.

1. **Substitute:**

$$\begin{aligned} u_n &= \sqrt{2 + u_{n-1}} \\ 2\cosh(\theta_n) &= \sqrt{2 + 2\cosh(\theta_{n-1})} \end{aligned}$$

2. **Apply Identity:**

$$\begin{aligned} 2\cosh(\theta_n) &= \sqrt{2(1 + \cosh(\theta_{n-1}))} \\ 2\cosh(\theta_n) &= \sqrt{2(2\cosh^2(\theta_{n-1}/2))} \\ 2\cosh(\theta_n) &= 2\cosh(\theta_{n-1}/2) \quad \text{(since } \cosh(x) > 0 \text{ for all } x\text{)} \end{aligned}$$

3. **Solve for θ_n:** This again gives $\theta_n = \theta_{n-1}/2$, so:

$$\theta_n = \frac{\theta_0}{2^n}$$

4. **Find the Closed-Form Solution:** From $u_0 = 2\cosh(\theta_0)$, we have $\theta_0 = \operatorname{arccosh}(u_0/2)$. The closed-form solution is:

$$u_n = 2\cosh\left(\frac{\operatorname{arccosh}(u_0/2)}{2^n}\right)$$

5. **Discuss Convergence:** As $n \to \infty$, the argument of cosh goes to zero:

$$\lim_{n\to\infty} u_n = 2\cosh(0) = 2 \cdot 1 = 2$$

Conclusion (Difference Equation): Both cases ($u_0 \leq 2$ and $u_0 > 2$) lead to the same limit, **2**, confirming our earlier result from the Monotone Convergence Theorem.

4.2 Complex Number Approach: Unified Solution

The two separate cases can be elegantly unified using complex numbers. Since $\cosh(x) = \cos(ix)$ for real x, we can express both cases using a single complex formulation.

4.2.1 Unified Complex Representation

We use the substitution $u_n = 2\cos(\theta_n)$, where θ_n is now allowed to be **complex**.

1. **Substitute:**

$$u_n = \sqrt{2 + u_{n-1}}$$
$$2\cos(\theta_n) = \sqrt{2 + 2\cos(\theta_{n-1})}$$

2. **Apply the Half-Angle Identity:** Using the identity $2\cos(x/2) = \sqrt{2 + 2\cos(x)}$ (which holds for complex x), we obtain:

$$2\cos(\theta_n) = 2\cos(\theta_{n-1}/2)$$

   This gives us the recurrence relation:

$$\theta_n = \frac{\theta_{n-1}}{2}$$

   Which has the solution:

$$\theta_n = \frac{\theta_0}{2^n}$$

3. **Determine θ_0 from u_0:**

   We need to solve $u_0 = 2\cos(\theta_0)$ for θ_0. The inverse function arccos can be extended to the complex plane.

   - **Case 1:** $0 < u_0 \leq 2$
     Here, $u_0/2 \in (0, 1]$, so we can take $\theta_0 = \arccos(u_0/2)$ where arccos returns a real value in $[0, \pi/2)$.
   - **Case 2:** $u_0 > 2$
     When $u_0/2 > 1$, we need to use the complex extension of arccos. Using the identity $\cos(i\alpha) = \cosh(\alpha)$ for real α, we can write:

$$\theta_0 = i\alpha, \quad \text{where } \alpha = \operatorname{arccosh}(u_0/2)$$

     This gives $u_0 = 2\cos(i\alpha) = 2\cosh(\alpha)$, which is consistent with our earlier hyperbolic substitution.

4. **Unified Closed-Form Solution:**

   The closed-form solution can be written as:

$$u_n = 2\cos\left(\frac{\theta_0}{2^n}\right)$$

   where θ_0 is determined by:

$$\theta_0 = \begin{cases} \arccos(u_0/2) & \text{if } 0 < u_0 \leq 2 \text{ (real } \theta_0) \\ i \cdot \operatorname{arccosh}(u_0/2) & \text{if } u_0 > 2 \text{ (purely imaginary } \theta_0) \end{cases}$$

5. **Complex Exponential Form:**

   Using Euler's formula, we can express this more elegantly. Since $\cos(z) = \frac{e^{iz}+e^{-iz}}{2}$ for any complex z, we have:

$$u_n = e^{i\theta_n} + e^{-i\theta_n} = e^{i\theta_0/2^n} + e^{-i\theta_0/2^n}$$

   where the principal square root ensures we take the real part.

   More precisely, since u_n must be real and positive:

$$u_n = \Re\left(e^{i\theta_0/2^n} + e^{-i\theta_0/2^n}\right) = 2\Re\left(\cos\left(\frac{\theta_0}{2^n}\right)\right)$$

6. **Verification of Both Cases:**

   - **When $u_0 \leq 2$:** θ_0 is real, so $u_n = 2\cos(\theta_0/2^n)$, matching our trigonometric solution.
   - **When $u_0 > 2$:** $\theta_0 = i\alpha$ where $\alpha = \operatorname{arccosh}(u_0/2)$ is real. Then:

$$u_n = 2\cos\left(\frac{i\alpha}{2^n}\right) = 2\cos\left(i \cdot \frac{\alpha}{2^n}\right) = 2\cosh\left(\frac{\alpha}{2^n}\right)$$

     which matches our hyperbolic solution.

7. **Convergence:**

   As $n \to \infty$, we have:

$$\lim_{n\to\infty} \frac{\theta_0}{2^n} = 0$$

   regardless of whether θ_0 is real or purely imaginary. Therefore:

$$\lim_{n\to\infty} u_n = 2\cos(0) = 2\cosh(0) = 2$$

4.2.2 Alternative Complex Exponential Derivation

We can also derive this directly from the recurrence relation using complex exponentials. Starting from:

$$u_n^2 = 2 + u_{n-1}$$

If we let $u_n = z_n + \bar{z}_n$ where z_n is complex and $\bar{z}_n$ is its complex conjugate, then $u_n = 2\Re(z_n)$. The recurrence becomes:

$$(z_n + \bar{z}_n)^2 = 2 + (z_{n-1} + \bar{z}_{n-1})$$

A particularly elegant choice is $z_n = e^{i\theta_n}$, which gives $u_n = 2\cos(\theta_n)$. Substituting into $u_n^2 = 2 + u_{n-1}$:

$$4\cos^2(\theta_n) = 2 + 2\cos(\theta_{n-1})$$

Using the double-angle identity $2\cos^2(x) = 1 + \cos(2x)$:

$$2(1 + \cos(2\theta_n)) = 2 + 2\cos(\theta_{n-1})$$

Simplifying gives $\cos(2\theta_n) = \cos(\theta_{n-1})$, which leads to $2\theta_n = \theta_{n-1}$ (choosing the appropriate branch), yielding $\theta_n = \theta_0/2^n$ as before.

Conclusion (Complex Approach): The complex number formulation elegantly unifies both the trigonometric and hyperbolic cases into a single expression $u_n = 2\cos(\theta_0/2^n)$, where θ_0 takes real or purely imaginary values depending on the initial condition. This demonstrates the power of complex analysis in providing unified solutions to problems that initially appear to require separate treatments.

4.3 Differential Equation Approach: Continuous Analogue

We can analyze the stability of the system by approximating the discrete difference equation with an autonomous differential equation. The discrete change is $u_n - u_{n-1} = \sqrt{2 + u_{n-1}} - u_{n-1}$.

We approximate this with a continuous function $u(t)$, where the change $u'(t)$ is analogous to $u_n - u_{n-1}$.

The associated differential equation is:

$$\frac{du}{dt} = \sqrt{2+u} - u \tag{2}$$

4.3.1 Finding Equilibria (Fixed Points)

The equilibria occur where the system is stable, i.e., $\frac{du}{dt} = 0$.

$$\begin{aligned} \sqrt{2+u} - u &= 0 \\ \sqrt{2+u} &= u \\ 2 + u &= u^2 \\ u^2 - u - 2 &= 0 \\ (u-2)(u+1) &= 0 \end{aligned}$$

Since $u_0 > 0$ (and all subsequent $u_n > 0$), we are only interested in non-negative equilibria. The only relevant equilibrium point is $u = 2$.

4.3.2 Analyzing Stability of the Equilibrium

We check the sign of $\frac{du}{dt}$ on either side of the equilibrium point $u = 2$. Let $g(u) = \frac{du}{dt} = \sqrt{2+u} - u$.

- **Region 1:** $0 < u < 2$
  Let's pick a test value, e.g., $u = 1$.

  $$g(1) = \sqrt{2+1} - 1 = \sqrt{3} - 1 \approx 1.732 - 1 = 0.732 > 0$$

  Since $\frac{du}{dt} > 0$, the function $u(t)$ is **increasing** in this region. The value of u moves *towards* 2.

- **Region 2:** $u > 2$
  Let's pick a test value, e.g., $u = 7$.

  $$g(7) = \sqrt{2+7} - 7 = \sqrt{9} - 7 = 3 - 7 = -4 < 0$$

  Since $\frac{du}{dt} < 0$, the function $u(t)$ is **decreasing** in this region. The value of u moves *towards* 2.

4.3.3 Conclusion (Differential Equation)

We can visualize this on a phase line:

$$\ldots(0) \xrightarrow{\text{increasing}} [2] \xleftarrow{\text{decreasing}} (\infty)\ldots$$

Because the system's flow (represented by the arrows) points towards $u = 2$ from both sides, $u = 2$ is an **asymptotically stable equilibrium**.

This analysis of the continuous analogue strongly implies that the discrete system (our sequence) will also be attracted to this fixed point. Regardless of the starting value $u_0 > 0$, the sequence will move towards and ultimately **converge to 2**.

5 Non-Monotonic Sequence with Noise

5.1 Problem Statement

What happens if we modify the recurrence relation by adding a small "noise" term? Consider the modified sequence:

$$u_n = \sqrt{2 + u_{n-1} + \epsilon_{n-1}}, \quad n \geq 1 \tag{3}$$

where (ϵ_n) is a sequence of non-negative numbers.

Question: Can we choose (ϵ_n) such that the sequence (u_n) becomes **non-monotonic** while still converging to 2?

This is an interesting extension that shows how sequences can converge even when they are not monotonic.

5.2 Solution

Yes! We can construct such a sequence. The key insight is to use a noise sequence (ϵ_n) that:

- Is large enough initially to "kick" the sequence above 2
- Decays to 0 as $n \to \infty$, so the modified sequence still converges to 2

5.2.1 The Epsilon Sequence

Let's define the noise sequence (ϵ_n) as a simple decaying function:

$$\epsilon_n = \frac{1}{n+1}$$

So, the sequence (ϵ_n) is:

- $\epsilon_0 = 1/1 = 1$
- $\epsilon_1 = 1/2 = 0.5$
- $\epsilon_2 = 1/3 \approx 0.333$
- $\epsilon_3 = 1/4 = 0.25$
- $\epsilon_4 = 1/5 = 0.2$
- ... (continuing indefinitely)

This sequence is simple, non-negative, and clearly converges to 0 as $n \to \infty$.

5.2.2 Analyzing the Modified Sequence

Now let's trace the sequence $u_n = \sqrt{2 + u_{n-1} + \epsilon_{n-1}}$ using this noise. The key to making u_n non-monotonic is to start it *below* 2, let the noise "kick" it *above* 2, and then watch it oscillate.

Let's pick an initial value of $u_0 = 1.5$.

1. **Calculate u_1:**

$$\begin{aligned} u_1 &= \sqrt{2 + u_0 + \epsilon_0} \\ u_1 &= \sqrt{2 + 1.5 + 1} = \sqrt{4.5} \\ u_1 &\approx 2.121 \end{aligned}$$

   The sequence increased: $2.121 > 1.5$

2. **Calculate u_2:**

$$\begin{aligned} u_2 &= \sqrt{2 + u_1 + \epsilon_1} \\ u_2 &= \sqrt{2 + 2.121 + 0.5} = \sqrt{4.621} \\ u_2 &\approx 2.149 \end{aligned}$$

   The sequence increased again: $2.149 > 2.121$

3. **Calculate u_3:**

$$\begin{aligned} u_3 &= \sqrt{2 + u_2 + \epsilon_2} \\ u_3 &= \sqrt{2 + 2.149 + 0.333} = \sqrt{4.482} \\ u_3 &\approx 2.117 \end{aligned}$$

   The sequence **decreased**: $2.117 < 2.149$

5.2.3 Conclusion

The sequence (u_n) is **non-monotonic** because it went up ($u_2 > u_1$) and then came down ($u_3 < u_2$).

However, the sequence **still converges to 2**. The "noise" $\epsilon_n = \frac{1}{n+1}$ gets smaller and smaller, so its effect diminishes. The sequence's natural "pull" towards the stable limit of 2 eventually takes over, and u_n will converge to 2.

This example demonstrates an important principle: **monotonicity is sufficient but not necessary for convergence**. Sequences can converge even when they oscillate, as long as the oscillations become smaller over time.

6 Final Thoughts

This problem beautifully illustrates several key concepts in sequence convergence:

- The Monotone Convergence Theorem as a powerful tool for proving convergence
- How to find potential limits by analyzing the recurrence relation

- The importance of considering different cases (initial conditions)
- That convergence does not require monotonicity

Understanding these ideas will serve you well in more advanced mathematical studies, including calculus, real analysis, and beyond.

7 Exercises

The following exercises will help you deepen your understanding of sequence convergence and the techniques we've discussed. Try to solve them using the methods from this document.

1. **Non-Monotonic Sequence with Alternating Noise**
   Consider the modified sequence:

   $$u_n = \sqrt{2 + u_{n-1} + \epsilon_{n-1}}, \quad n \geq 1$$

   where $\epsilon_n = (-1)^n \cdot \frac{1}{n+1}$ for $n \geq 0$ and $u_0 = 1.5$. Show that this sequence converges to 2 despite being non-monotonic.

   Hint: The noise alternates in sign. Consider the behavior of u_n when the noise is positive versus negative. Use the fact that $|\epsilon_n| \to 0$ as $n \to \infty$.

2. **Modified Recurrence:** $\sqrt{1 + u_{n-1}}$
   Consider the sequence defined by:

   $$u_n = \sqrt{1 + u_{n-1}}, \quad n \geq 1$$

   with $u_0 > 0$. Determine whether this sequence converges, and if so, find its limit. Prove your result using the Monotone Convergence Theorem.

   Hint: Start by finding the potential fixed point(s) by solving $L = \sqrt{1 + L}$ for L. Then analyze monotonicity and boundedness similar to the original problem.

3. **Cube Root Recurrence:** $\sqrt[3]{6 + u_{n-1}}$
   Consider the sequence defined by:

   $$u_n = \sqrt[3]{6 + u_{n-1}}, \quad n \geq 1$$

   with $u_0 > 0$.

   i) Show that the associated differential equation $\frac{du}{dt} = \sqrt[3]{6 + u} - u$ has exactly one fixed point.
   ii) Prove that the sequence (u_n) is monotonic (either increasing or decreasing depending on u_0).
   iii) Find the limit of the sequence and prove convergence.

Hint: For part (i), solve $\sqrt[3]{6+u} = u$ to find the fixed point. For part (ii), compare u_n with u_{n-1} by considering the function $f(x) = \sqrt[3]{6+x} - x$.

4. **Generalized Recurrence:** $\sqrt{a + u_{n-1}}$
   Consider the sequence defined by:

$$u_n = \sqrt{a + u_{n-1}}, \quad n \geq 1$$

   where $a > 0$ is a constant and $u_0 > 0$.

   i) Find all possible limits of the sequence in terms of a.
   ii) For which values of a does the sequence converge for any initial value $u_0 > 0$?
   iii) Investigate the behavior when $a = 0$. Does the sequence converge? If so, to what?

Hint: Start by finding fixed points: solve $L = \sqrt{a + L}$. Consider the discriminant of the resulting quadratic equation. For part (iii), think carefully about the domain and what happens when $a = 0$.

5. **Rate of Convergence**
   For the original sequence $u_n = \sqrt{2 + u_{n-1}}$ with $0 < u_0 < 2$, we showed that $u_n \to 2$ as $n \to \infty$.

   i) Show that $2 - u_n = O(1/2^n)$ as $n \to \infty$ (i.e., the error decreases exponentially).
   ii) More precisely, using the closed-form solution $u_n = 2\cos(\theta_0/2^n)$, show that:

$$2 - u_n \sim \frac{\theta_0^2}{2^{2n+1}} \quad \text{as } n \to \infty$$

   where $\theta_0 = \arccos(u_0/2)$ and $\sim$ means asymptotic equivalence.

Hint: For part (i), use the fact that $u_n = 2\cos(\theta_0/2^n)$ and consider the Taylor expansion of $\cos(x)$ near $x = 0$. For part (ii), use $\cos(x) = 1 - x^2/2 + O(x^4)$.

8 Solutions to Exercises

8.1 Solution to Exercise 1: Non-Monotonic Sequence with Alternating Noise

We need to show that the sequence $u_n = \sqrt{2 + u_{n-1} + \epsilon_{n-1}}$ with $\epsilon_n = (-1)^n \cdot \frac{1}{n+1}$ and $u_0 = 1.5$ converges to 2, despite being non-monotonic.

Step 1: Showing the sequence is non-monotonic

Let's compute the first few terms:

$$\begin{aligned}
u_0 &= 1.5 \\
u_1 &= \sqrt{2 + 1.5 + \epsilon_0} = \sqrt{2 + 1.5 + 1} = \sqrt{4.5} \approx 2.121 \\
u_2 &= \sqrt{2 + 2.121 + \epsilon_1} = \sqrt{2 + 2.121 - 0.5} = \sqrt{3.621} \approx 1.903 \\
u_3 &= \sqrt{2 + 1.903 + \epsilon_2} = \sqrt{2 + 1.903 + 0.333} = \sqrt{4.236} \approx 2.058
\end{aligned}$$

We see that $u_1 \approx 2.121$, $u_2 \approx 1.903$, and $u_3 \approx 2.058$. So $u_1 > u_2$ (the sequence decreases from u_1 to u_2) and $u_2 < u_3$ (the sequence increases from u_2 to u_3), showing the sequence is **non-monotonic**.

Step 2: Showing convergence to 2

We have $u_n = \sqrt{2 + u_{n-1} + \epsilon_{n-1}}$ where $|\epsilon_n| = \frac{1}{n+1} \to 0$ as $n \to \infty$.

Since $|\epsilon_n| \to 0$, for any $\delta > 0$, there exists N such that $|\epsilon_n| < \delta$ for all $n \geq N$. This means that for large n, the modified sequence behaves almost like the original sequence $v_n = \sqrt{2 + v_{n-1}}$, which we know converges to 2.

More rigorously, consider the difference:

$$|u_n - 2| = \left|\sqrt{2 + u_{n-1} + \epsilon_{n-1}} - 2\right|$$

Using the fact that $\sqrt{2 + u_{n-1} + \epsilon_{n-1}} = \sqrt{2 + u_{n-1}} \cdot \sqrt{1 + \frac{\epsilon_{n-1}}{2+u_{n-1}}}$, and since $u_n > 0$ for all n, we have:

$$|u_n - 2| \leq \left|\sqrt{2 + u_{n-1}} - 2\right| + \left|\sqrt{2 + u_{n-1}}\right| \cdot \left|\sqrt{1 + \frac{\epsilon_{n-1}}{2 + u_{n-1}}} - 1\right|$$

Since $|\epsilon_{n-1}| \to 0$ and u_{n-1} is bounded (the sequence oscillates around 2), the second term goes to 0. The first term represents the error from the unmodified sequence, which also converges to 0.

By the Squeeze Theorem (or using the fact that the "noise" decays to 0), we conclude that $\lim_{n\to\infty} u_n = 2$.

8.2 Solution to Exercise 2: Modified Recurrence $\sqrt{1 + u_{n-1}}$

Consider $u_n = \sqrt{1 + u_{n-1}}$ with $u_0 > 0$.

Step 1: Finding potential limits

If the sequence converges to L, then:

$$L = \sqrt{1 + L} \quad \Rightarrow \quad L^2 = 1 + L \quad \Rightarrow \quad L^2 - L - 1 = 0$$

Using the quadratic formula:

$$L = \frac{1 \pm \sqrt{1 + 4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

Since $u_n > 0$ for all n, we must have $L \geq 0$. The negative root $\frac{1-\sqrt{5}}{2} < 0$ is invalid. Therefore, the only possible limit is:

$$L = \frac{1+\sqrt{5}}{2} = \phi$$

where ϕ is the golden ratio (approximately 1.618).

Step 2: Proving convergence using Monotone Convergence Theorem

We analyze monotonicity by comparing u_n and u_{n-1}:

$$u_n > u_{n-1} \quad \Leftrightarrow \quad \sqrt{1+u_{n-1}} > u_{n-1} \quad \Leftrightarrow \quad 1 + u_{n-1} > u_{n-1}^2$$

This simplifies to:

$$u_{n-1}^2 - u_{n-1} - 1 < 0$$

The roots of $x^2 - x - 1 = 0$ are $\frac{1\pm\sqrt{5}}{2}$. The quadratic is negative between these roots. Since we only care about positive values:

$$0 < u_{n-1} < \frac{1+\sqrt{5}}{2} = \phi$$

So:

- If $u_{n-1} < \phi$, then $u_n > u_{n-1}$ (sequence increases)
- If $u_{n-1} > \phi$, then $u_n < u_{n-1}$ (sequence decreases)
- If $u_{n-1} = \phi$, then $u_n = \phi$ (sequence is constant)

Case 1: $0 < u_0 < \phi$ By induction, $u_n < \phi$ for all n (since $\sqrt{1+u_{n-1}} < \sqrt{1+\phi} = \phi$). The sequence is increasing and bounded above by ϕ, so it converges. The limit must be ϕ.

Case 2: $u_0 > \phi$ By induction, $u_n > \phi$ for all n. The sequence is decreasing and bounded below by ϕ, so it converges. The limit must be ϕ.

Case 3: $u_0 = \phi$ Then $u_n = \phi$ for all n, so it trivially converges to ϕ.

Conclusion: The sequence converges to $\phi = \frac{1+\sqrt{5}}{2}$ for any $u_0 > 0$.

8.3 Solution to Exercise 3: Cube Root Recurrence $\sqrt[3]{6+u_{n-1}}$

Consider $u_n = \sqrt[3]{6+u_{n-1}}$ with $u_0 > 0$.

Part (i): Fixed point of the differential equation

The associated differential equation is $\frac{du}{dt} = \sqrt[3]{6+u} - u$. Fixed points occur when:

$$\sqrt[3]{6+u} - u = 0 \quad \Rightarrow \quad \sqrt[3]{6+u} = u \quad \Rightarrow \quad 6 + u = u^3$$

Rearranging: $u^3 - u - 6 = 0$.

Let $f(u) = u^3 - u - 6$. We have $f(0) = -6 < 0$ and $f(2) = 8 - 2 - 6 = 0$. Actually, $f(2) = 0$, so $u = 2$ is a root. Let's verify: $f(2) = 8 - 2 - 6 = 0$. Yes, $u = 2$ is a fixed point.

To show this is the only positive fixed point, we can factor the cubic: $u^3 - u - 6 = (u-2)(u^2+2u+3)$. The quadratic u^2+2u+3 has discriminant $4-12 = -8 < 0$, so it has no real roots. Therefore, $u = 2$ is the only real root, and thus the differential equation has **exactly one fixed point**.

Part (ii): Monotonicity

We compare u_n and u_{n-1}:

$$u_n \gtreqless u_{n-1} \quad \Leftrightarrow \quad \sqrt[3]{6+u_{n-1}} \gtreqless u_{n-1} \quad \Leftrightarrow \quad 6+u_{n-1} \gtreqless u_{n-1}^3$$

This gives: $u_{n-1}^3 - u_{n-1} - 6 \lesseqgtr 0$.

Since $f(x) = x^3 - x - 6 = (x-2)(x^2+2x+3)$ and $x^2+2x+3 > 0$ for all real x, the sign of $f(x)$ is determined by $(x-2)$. Therefore:

- If $u_{n-1} < 2$, then $f(u_{n-1}) < 0$, so $u_n > u_{n-1}$ (sequence increases)
- If $u_{n-1} > 2$, then $f(u_{n-1}) > 0$, so $u_n < u_{n-1}$ (sequence decreases)
- If $u_{n-1} = 2$, then $u_n = 2$ (sequence is constant)

Therefore, the sequence is **monotonic** (either strictly increasing if $u_0 < 2$, strictly decreasing if $u_0 > 2$, or constant if $u_0 = 2$).

Part (iii): Finding the limit and proving convergence

By induction:

- If $u_0 < 2$, then $u_n < 2$ for all n, and the sequence is increasing and bounded above by 2.
- If $u_0 > 2$, then $u_n > 2$ for all n, and the sequence is decreasing and bounded below by 2.

By the Monotone Convergence Theorem, the sequence converges. The limit must satisfy $L = \sqrt[3]{6+L}$, which gives $L^3 - L - 6 = 0$. Since $L = 2$ is the unique positive root, we have $\lim_{n\to\infty} u_n = 2$.

8.4 Solution to Exercise 4: Generalized Recurrence $\sqrt{a+u_{n-1}}$

Consider $u_n = \sqrt{a+u_{n-1}}$ where $a > 0$ is a constant and $u_0 > 0$.

Part (i): Finding all possible limits

If the sequence converges to L, then:

$$L = \sqrt{a+L} \quad \Rightarrow \quad L^2 = a + L \quad \Rightarrow \quad L^2 - L - a = 0$$

Using the quadratic formula:

$$L = \frac{1 \pm \sqrt{1+4a}}{2}$$

Since $u_n > 0$ for all n, we require $L \geq 0$. The negative root $\frac{1-\sqrt{1+4a}}{2}$ is negative for $a > 0$, so it's invalid. Therefore, the only possible limit is:

$$L = \frac{1+\sqrt{1+4a}}{2}$$

Part (ii): Conditions for convergence

For the sequence to converge for any $u_0 > 0$, we need to ensure it's monotonic and bounded. The analysis is similar to the original problem:

$u_n > u_{n-1}$ when $\sqrt{a+u_{n-1}} > u_{n-1}$, which gives $u_{n-1}^2 - u_{n-1} - a < 0$.

The roots of $x^2 - x - a = 0$ are $\frac{1\pm\sqrt{1+4a}}{2}$. For positive values, the quadratic is negative when:

$$0 < u_{n-1} < \frac{1+\sqrt{1+4a}}{2} = L$$

So:

- If $u_{n-1} < L$, then $u_n > u_{n-1}$ (increasing)
- If $u_{n-1} > L$, then $u_n < u_{n-1}$ (decreasing)

By induction:

- If $0 < u_0 < L$, then $u_n < L$ for all n, and the sequence is increasing and bounded above.
- If $u_0 > L$, then $u_n > L$ for all n, and the sequence is decreasing and bounded below.

Therefore, the sequence **always converges** to $L = \frac{1+\sqrt{1+4a}}{2}$ for any $a > 0$ and any $u_0 > 0$.

Part (iii): Special case $a = 0$

When $a = 0$, the recurrence becomes $u_n = \sqrt{u_{n-1}} = u_{n-1}^{1/2}$.

If $u_0 > 0$, then by induction $u_n > 0$ for all n. We have:

$$u_n = u_{n-1}^{1/2} = u_0^{1/2^n}$$

As $n \to \infty$, we have $1/2^n \to 0$, so:

$$\lim_{n\to\infty} u_n = u_0^0 = 1$$

(Note: This is valid since $u_0 > 0$.)

Therefore, when $a = 0$, the sequence converges to **1** for any $u_0 > 0$.

8.5 Solution to Exercise 5: Rate of Convergence

For the original sequence $u_n = \sqrt{2 + u_{n-1}}$ with $0 < u_0 < 2$, we have the closed-form solution:

$$u_n = 2\cos\left(\frac{\theta_0}{2^n}\right)$$

where $\theta_0 = \arccos(u_0/2) \in (0, \pi/2)$.

Part (i): Showing $2 - u_n = O(1/2^n)$

We have:

$$2 - u_n = 2 - 2\cos\left(\frac{\theta_0}{2^n}\right) = 2\left(1 - \cos\left(\frac{\theta_0}{2^n}\right)\right)$$

Using the identity $1 - \cos(x) = 2\sin^2(x/2)$:

$$2 - u_n = 2 \cdot 2\sin^2\left(\frac{\theta_0}{2^{n+1}}\right) = 4\sin^2\left(\frac{\theta_0}{2^{n+1}}\right)$$

For small x, we have $\sin(x) \approx x$, so:

$$2 - u_n = 4\sin^2\left(\frac{\theta_0}{2^{n+1}}\right) \approx 4\left(\frac{\theta_0}{2^{n+1}}\right)^2 = \frac{\theta_0^2}{2^{2n}}$$

More precisely, since $|\sin(x)| \leq |x|$ for all x:

$$2 - u_n = 4\sin^2\left(\frac{\theta_0}{2^{n+1}}\right) \leq 4\left(\frac{\theta_0}{2^{n+1}}\right)^2 = \frac{\theta_0^2}{2^{2n}}$$

This shows that $2 - u_n = O(1/2^{2n}) = O(1/4^n)$, which is even better than $O(1/2^n)$.
Part (ii): Asymptotic equivalence
Using the Taylor expansion $\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$, we have:

$$\cos\left(\frac{\theta_0}{2^n}\right) = 1 - \frac{\theta_0^2}{2^{2n+1}} + O\left(\frac{\theta_0^4}{2^{4n}}\right)$$

Therefore:

$$u_n = 2\cos\left(\frac{\theta_0}{2^n}\right) = 2 - \frac{\theta_0^2}{2^{2n}} + O\left(\frac{\theta_0^4}{2^{4n}}\right)$$

So:

$$2 - u_n = \frac{\theta_0^2}{2^{2n}} + O\left(\frac{\theta_0^4}{2^{4n}}\right) = \frac{\theta_0^2}{2^{2n}}\left(1 + O\left(\frac{\theta_0^2}{2^{2n}}\right)\right)$$

Taking the limit:

$$\lim_{n\to\infty} \frac{2 - u_n}{\theta_0^2/2^{2n}} = \lim_{n\to\infty}\left(1 + O\left(\frac{\theta_0^2}{2^{2n}}\right)\right) = 1$$

This means:

$$2 - u_n \sim \frac{\theta_0^2}{2^{2n}} \quad \text{as } n \to \infty$$

Or equivalently:

$$2 - u_n \sim \frac{\theta_0^2}{2^{2n+1}} \cdot 2 = \frac{\theta_0^2}{2^{2n+1}} \cdot \frac{2^{2n+1}}{2^{2n}} = \frac{\theta_0^2}{2^{2n}}$$

Wait, let me recalculate more carefully:

$$2 - u_n = 2\left(1 - \cos\left(\frac{\theta_0}{2^n}\right)\right) = 2 \cdot \frac{\theta_0^2}{2^{2n+1}} + O\left(\frac{\theta_0^4}{2^{4n}}\right) = \frac{\theta_0^2}{2^{2n}} + O\left(\frac{\theta_0^4}{2^{4n}}\right)$$

So $2 - u_n \sim \frac{\theta_0^2}{2^{2n}}$ as $n \to \infty$.

The formula in the exercise statement has 2^{2n+1} in the denominator, which would give $\frac{\theta_0^2}{2^{2n+1}}$. Let me check: if we want $\frac{\theta_0^2}{2^{2n+1}}$, then:

$$\frac{2 - u_n}{\theta_0^2/2^{2n+1}} = \frac{2^{2n+1}(2 - u_n)}{\theta_0^2} = \frac{2^{2n+1} \cdot \frac{\theta_0^2}{2^{2n}}}{\theta_0^2} = 2$$

So actually $2 - u_n \sim 2 \cdot \frac{\theta_0^2}{2^{2n+1}} = \frac{\theta_0^2}{2^{2n}}$. The statement in the exercise appears to have a slight error, but the asymptotic behavior is correctly described: the error decreases like $\theta_0^2/2^{2n}$.

For completeness, let's verify the statement as given:

$$2 - u_n = \frac{\theta_0^2}{2^{2n}} + O(2^{-4n}) = \frac{\theta_0^2}{2^{2n+1}} \cdot 2 + O(2^{-4n})$$

More precisely, we have:

$$2 - u_n = \frac{\theta_0^2}{2^{2n}} + O\left(\frac{\theta_0^4}{2^{4n}}\right) = \frac{\theta_0^2}{2^{2n+1}} \cdot 2 + O\left(\frac{\theta_0^4}{2^{4n}}\right)$$

The dominant term is $\frac{\theta_0^2}{2^{2n}}$, which can be written as $\frac{\theta_0^2}{2^{2n+1}} \cdot 2$. While the exercise statement uses $\sim \frac{\theta_0^2}{2^{2n+1}}$, the precise asymptotic is $2 - u_n \sim \frac{\theta_0^2}{2^{2n}}$. Both expressions correctly capture the exponential decay rate, differing only by a constant factor.

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